

SUMMARY OF SPATIO-TEMPORAL MODELLING OF THE MOUSE BRAIN BY JAMES ZOU GROUP

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1. WHAT THE PAPER DOES

1.1. Notations. There are n_{samples} samples, $n_{\text{cell types}}$ cell types, and n_{genes} genes in total. We use $r \in [n_{\text{samples}}]$ to index the sample and $q \in [n_{\text{cell types}}]$ to index the cell types. For each sample of interest, there are $n_{\text{cells}}^{q,r}$ cells. On each cell, each of the n_{genes} genes is either expressed or not expressed.

1.2. SpatialSmooth. Spatial Smooth is a way to regularise the gene expression matrix.

Fix a sample r and a cell type q . We surpress the index q, r for notational simplicity. For a given cell of the cell type in the sample, consider its $k(=20)$ -nearest neighbours in the same cell type. The L_1 -normalised adjacency mamtrix is given by $S \in \mathbb{B}^{n_{\text{cells}} \times n_{\text{cells}}}$, $\mathbb{B} = \{0, 1\}$. Let $X \in \mathbb{R}_+^{n_{\text{cells}} \times n_{\text{genes}}}$ be the (log-normalised) gene expression matrix, where genes are columns and cells are rows.

Let $\alpha = .8$ be the smoothing parameter. We then compute a collection of progressively smoothed gene expression matrices using

$$X_t = \alpha S X_{t-1} + (1 - \alpha) X, \quad \text{for } t \in [30].$$

If at some point the iteration has converged (passing some pre-set threshold) then it is stopped early. Otherwise, it is stopped after 30 rounds. This yields a smoothed gene expression matrix X_{final} . Standardise X_{final} to obtain $\tilde{X}$.

At this point, $\tilde{X}$ contains the gene expression information mediated through the spatial structure of the cells. This is the only point in this model where spatial information is used in the construction of the spatial aging clock in the subsequent section – no dependence on spatial structure comes from anything beside the dependence on $\tilde{X}$.

This procedure generates one such $\tilde{X}$ for each cell type and each sample, so we have $\tilde{X}^{1,1}, \dots, \tilde{X}^{n_{\text{cell types}} \times n_{\text{samples}}}$, where $\tilde{X}^{q,r} \in \mathbb{R}_+^{n_{\text{cells}}^{q,r} \times n_{\text{genes}}^{q,r}}$.

1.3. Spatial Aging Clock. Spatial Aging Clock is a regression that predicts the age of the cell using the gene expressions.

First, compute $\tilde{X}^{1,1}, \dots, \tilde{X}^{n_{\text{cell types}} \times n_{\text{samples}}}$ as above.

Second, construct the response vectors $Y^{q,r} \in \mathbb{R}^{n_{\text{cells}}^{q,r}}$, which contains the ages of the cells coming from the samples. Since each sample comes from one mouse, all entries in the vector $Y^{q,r}$ are the same for each fixed r . There are around 20 possible values for the age variable in total.

Third, for each fixed cell type q , we combine the corresponding responses and features $(\tilde{X}^{q,r}, Y^{q,r})$ across different values of r . This enables us to perform a LASSO algorithm (cv=5, n.alphas=20, max_iter=10000) that regresses all of $Y^{q,r}$ responses on all of $\tilde{X}^{q,r}$

features. For cell type q , the genes relevant for age prediction are chosen. The data size used for LASSO for cell type q is $\sum_{r=1}^{n_{\text{samples}}} n_{\text{cells}}^{q,r}$.

The upshot of this is a formula for predicting the age of a cell (of cell type q and from sample r) given the values in its own row in the matrix $\tilde{X}^{q,r}$, which of course gives different predictions for different cells within the same sample – even though in the training data we know that all cells within the sample r are of the same age.

1.4. Spatial Aging Clock Measures Aging. Once we have validated that this spatial aging clock has good performance, we can use its predictions as a measure of aging. This allows us to have a quantification of *how much the mouse has aged*, which is distinct from how many calendar years it has spent on Earth. For example, mice 1 and 2 have both lived on Earth for 24 months, but perhaps the spatial aging clock predicts all of the cells of mouse 1 to be older than that of mouse 2. This can happen if we give mouse 1 treatments that induce aging while give mouse 2 all the youthful / rejuvenating treatments.

1.5. Cell Proximity Effects: Effects of Spatial Proximity to Effector Cells on Aging Acceleration of the Target Cells. We can do cell level analysis of aging this way – one particular cell may have aged more than another cell (of the same sample), according to the Spatial Aging Clock.

Specifically, fix a target cell type q_T and an effector cell type q_E . Some target cells are near to the set of effector cells. Some target cells are far from the set of effector cells. We can now compute Cohen’s d measure of effect size between the age acceleration of the Near target cells compared to those of the Far target cells – positive proximity effect values indicate a pro-aging effect exerted by the q_E cell type on the q_T cell type.

This allows the identification of pro-aging and pro-rejuvenating cell types.

1.6. Spatial Permutation Analysis. To show that the spatial structure the above section, the spatial coordinates of the cells are randomly permuted. When this step is done, all of the proximity effects observed in the preceding section disappear indeed.

1.7. Spatial Gradation of Cell Proximity Effects. Cell proximity effects can be computed as a function of the unit distance between effector and target cells.

1.8. What I Don’t Know.

- (1) Transcriptomics spillover effects.
- (2) What kind of math can be done with this kind of method?

2. WHAT WE CAN DO WITH IT

- (1) The copy-and-paste thing to do is to change the “age in calendar years” in this work to “egg health / atresia level of the follicle” in Hattie data.
- (2) We have been fitting the covariance model on gene expression levels directly thus far. This is unsupervised learning. This has proved to be challenging computationally. They are using the spatial information somehow indirectly – first in the identification of k -nearest neighbours in Spatial Smooth and second in the are predicting 20 possible values in a way that somehow incorporates the spatial structure. This is supervised learning. Perhaps, we can formulate a supervised learning problem on our data set. That can perhaps make the computations easier.

- (3) We can redo our model fitting to be within cell types, since that sounds like a reasonable thing to do and is something this work did.
- (4) Where we are taking this to, I think, depends on the goal of the project. There are two possibilities.
 - (a) To answer the scientific question of Hattie, possibly using existing statistical methods. In this case, we can do the Copy-and-Paste strategy described above. We can be more sophisticated in our analysis of the empirical variogram. We can do the hypothesis tests of correlation decay over spatial range. There seem to be several computationally simple things to do that can give Hattie something that she seems to need.
 - (b) To develop a new general-purpose statistical method / pipeline for spatially resolved genes expression level data sets. There are many great things about having a parametric model with interpretable parameters...

If our goal is to develop some method that has some math behind it and can be claimed to be universal in some sense, then we perhaps need to keep on with the parametric fitting; on the contrary, perhaps more meaningful methods can be developed with the specification using the dataset and the scientific problem at hand.

There are some other things that we can do with the existing model.

- (1) Try PCA and testing for Gaussianity.
- (2) Compare with baseline model of Joe Guinness again. Is it even better, if our initialisations are different from ground truth?
- (3) Record the time complexity of the cross terms fitting codes as a function of number of locations and number of features. At least we should know for how many locations this model is useful for.

3. CHATGPT'S INTRODUCTION TO COHEN'S d

(This reads like a subtler version of average treatment effect.)

Cohen's d is a statistical measure that quantifies the difference between two group means in terms of standard deviation units. It is commonly used to measure the *effect size*, which indicates the practical significance of differences, independent of sample size.

3.1. **Formula.** The formula for Cohen's d is:

$$d = \frac{M_1 - M_2}{s}$$

where:

- M_1 and M_2 are the means of the two groups,
- s is the pooled standard deviation, calculated as:

$$s = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Here:

- n_1 and n_2 are the sample sizes of the two groups,
- s_1^2 and s_2^2 are the variances of the two groups.

3.2. **Interpretation.** The values of d are interpreted as follows:

- **Small effect:** $d = 0.2$,
- **Medium effect:** $d = 0.5$,
- **Large effect:** $d = 0.8$ or more.

These are general guidelines, and the context of the research determines what constitutes a meaningful effect size.

3.3. **Example.** Suppose you conduct an experiment comparing two groups:

- Group 1 (new technique): Mean $M_1 = 85$, standard deviation $s_1 = 10$, and $n_1 = 30$,
- Group 2 (standard technique): Mean $M_2 = 75$, standard deviation $s_2 = 15$, and $n_2 = 30$.

First, calculate the pooled standard deviation:

$$s = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Substitute the values:

$$s = \sqrt{\frac{(30 - 1)(10^2) + (30 - 1)(15^2)}{30 + 30 - 2}} = \sqrt{\frac{29 \cdot 100 + 29 \cdot 225}{58}} = \sqrt{\frac{2900 + 6525}{58}} = 12.91$$

Next, compute Cohen's d :

$$d = \frac{M_1 - M_2}{s} = \frac{85 - 75}{12.91} = 0.775$$

3.4. **Conclusion.** The effect size $d = 0.775$ indicates a **large effect**, suggesting that the new technique has a substantial impact on scores compared to the standard technique.

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